# Linear regression:

## assumptions

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# Multicollinearity

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Multicollinearity occurs when two or more explanatory variables are correlated (in any form) could cause that coefficient change of sign and its standard error increase (harder make inference, a significant variable could appear as not)

**VIF (Variance Inflation Factor)**

Diagnose **linear** multicollinearity.

# VIF

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Idea is how much icreases variance of coefficient given the correlation of predictors variables.

**...**

**VIF** is calculated by each predictor in the model

$$VIF(x_i) = 1 - R^2 \tag{1}$$

$R^2$ is obtained from a linear regression where $x_i$ is dependet of all another variables.

$$x_i = \sum_{j \neq i}^{k} \beta' x_j + u$$

# Rule of thumb in VIF

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- **VIF = 1** no multicollinearity
- **VIF** $>$ **1** and **VIF** $<=$ **5** acceptable
- **VIF** $>$ **5** strong multicollinearity

# fix multicollinearty

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- combine variables
- remove one
- regularization
- dimensionality reduction

# Breush-pagan test

used to test heteroskedasticity in linear regression.

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Test wether the variance of the errors in a linear regression are dependent on the values of explanatory variables (heteroskesdasticity)

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$$\begin{aligned} H_0 &: Homocedasticity \\ H_1 &: Heterocedaskticity \end{aligned} \tag{2}$$

...

test statistics is $\chi^2$ distribution, if the test statistics has associated a $p - value < 0.05$ the reject $H_0$ assuming heterocedasticity.

# fix heterocedaskticity

**...**

- Transform variables (especially the dependent) uses logarithms, or square root.
- Box-cox transformation
- Another methods..

# Independence

**Breusch-Godfrey Test**

$$\begin{aligned} H_0 &: \textbf{independent residuals} \\ H_1 &: \textbf{correlated residuals} \end{aligned} \tag{3}$$

if $p - value < 0.05$ reject the null hypothesis, assuming correlated residuals.